Exercise Sheet 1

Exercise 1. $\pi(x) \ge \log\log x$. pf: $p_{n+1} \le p_1 \cdots p_n + 1$，归纳，$p_n \le 2^{2^{n-1}} \le e^{e^{n-1}}$，取对数。

Exercise 2. (a) $\tau(n^2) = \sum_{d|n} \mu(d)\tau(n/d)^2$; (b) $\sum_{d^k|n} \mu(d) = 1_{\text{无} k \text{ 次方因子}}$; (c) $\mu(n) = \sum_{1\le a\le n,(a,n)=1} e^{2\pi i a/n}$. (a) pf: $\tau(\cdot^2) = \mu * \tau(\cdot)^2$，只需 $\tau(\cdot)^2 = \tau(\cdot^2) * u$，显。(b) pf: 最大 k 次方因子设为 $p_1^{k\alpha_1}\cdots p_m^{k\alpha_m}$，若为 1，则 LHS= 1，否则为 $\sum_{d|p_1\cdots p_m} \mu(d) = 0$。(c) pf: 先证 RHS 可乘，因为 m, n 互素时，在模 mn 意义下 $\{am+bn \mid a \in (\mathbb{Z}/n)^\times,\ b \in (\mathbb{Z}/m)^\times\} = (\mathbb{Z}/mn)^\times$，再说 p^α 时 LHS=RHS（用 p 进制）。

Exercise 3. (a) $\sum_{d|n} \mu(d)^2 = 2^{\omega(n)}$; (b) $\sum_{d|n} \mu(d)\tau(d) = (-1)^{\omega(n)}$. (a) pf: 显。(b) pf: LHS= $\sum_{k=0}^{\omega(n)} \binom{\omega(n)}{k}(-1)^k 2^k = (1-2)^{\omega(n)} = (-1)^{\omega(n)}$。

Exercise 4. f 可乘，则 f^{-1} 完全可乘 iff $f(p^\ell) = 0$，$\ell \ge 2$. pf: 已知可乘则完全可乘等价于 $f^{-1} = \mu f$ 故显。

Exercise 5. $\{f : f * f = u\}$ 恰有两个，是否可乘? pf: f 满足 $f(1) = \pm 1$，递推 $f(n) = \frac{1}{2f(1)}(1 - \sum_{d|n,\, d\ne 1,n} f(d)f(n/d))$，该递推确定，故恰有两个。若 $f(1) = -1$ 不可乘; $f(1) = 1$ 可乘（用反证法找最小的 mn）。

Exercise 6. $(f * g)^{-1} = f^{-1} * g^{-1}$. 显。

Exercise 7. $\operatorname{supp}\Lambda_k \subseteq \{n : \omega(n) \le k\}$; $0 \le \Lambda_k(n) \le (\log n)^k$. ① pf: $\Lambda_k = \Lambda_{k-1}\log + \Lambda_{k-1} * \Lambda$，故显。② $\Lambda \ge 0$，是因为递推式，“$\le$”是因为 $\log^k = \Lambda_k * u \Rightarrow \Lambda_k(n) = \log^k n - \sum_{d|n,\, d<n} \Lambda_k(d) \le \log^k n$。

Exercise Sheet 2

Exercise 1. 求 $f \in A$ s.t. $\frac{1}{\varphi(n)} = \sum_{d|n} \frac{1}{d} f(n/d)$. pf: $a(n) = 1/n$ 有 $a^{-1}(n) = \mu(n)/n$，故 $f = \frac{1}{\varphi} * a^{-1}$ 可乘，$f(p) = \frac{1}{p(p-1)}$，$f(p^\alpha) = 0$ $(\alpha \ge 2)$，故 $f(n) = \mu(n)^2 \prod_{p|n} \frac{1}{p(p-1)} = \frac{\mu(n)^2}{n\varphi(n)}$。

Exercise 2. $\sum_{n\le x} \frac{1}{\varphi(n)} = C_1 \log x + C_2 + O(\log x/x)$. pf: $\sum_{n\le x} \frac{1}{\varphi(n)} = \sum_{n\le x} a * f(n) = \sum_{d\le x} f(d) \sum_{m\le x/d} \frac{1}{m} = \sum_{d\le x} f(d)(\log\frac{x}{d} + \gamma + O(d/x)) := \log x \sum_{d\ge 1} f(d) + (-\sum_{d\ge1} f(d)\log d + \gamma\sum_{d\ge1} f(d)) - \log x \sum_{d>x} f(d) - (-\sum_{d>x} f(d)\log d + \gamma \sum_{d>x} f(d)) + O(\sum_{d\le x} f(d)d/x)$。由以下估计可证原题: ① $\sum_{d\ge1} f(d)$, $\sum_{d\ge1} f(d)\log d$ 收敛。pf: $\varphi(n) \gg n/\log\log n$（由此也可知 $\sum_{n\le x} 1/\varphi(n) = O(\log x)$）。② $\sum_{d\le x} f(d)d = O(\log x)$。pf: $|\sum_{d\le x} f(d)d| = |\sum_{d\le x} \mu(d)^2/\varphi(d)| \le |\sum_{d\le x} 1/\varphi(d)| = O(\log x)$。③ $\sum_{d>x} f(d) = O(1/x)$。pf: $f(d) = d^{-2}\mu(d)^2 d/\varphi(d)$，记为 $d^{-2}b(d)$，由 2-1，$\frac{n}{\varphi(n)} = n\, a * f(n) = n\sum_{d|n} \frac{d\mu^2(d)}{n\, d\varphi(d)} = \sum_{d|n} \frac{\mu^2(d)}{\varphi(d)}$，即 $|b(n)| \le n/\varphi(n) = \sum_{d|n} \mu^2(d)/\varphi(d)$，那 $|\sum_{n\le x} b(n)| \le \sum_{n\le x}\sum_{d|n} \mu^2(d)/\varphi(d) = \sum_{d\le x} \frac{\mu^2(d)}{\varphi(d)}[x/d] \le x\sum_{d\le x} \frac{\mu^2(d)}{d\varphi(d)} = x\sum_{d\le x} f(d) = O(x)$，即 $\sum_{d>x} f(d) = \lim_{N\to\infty} \sum_{x<d\le N} \frac{b(d)}{d^2} = \lim_{N\to\infty} \int_x^N \frac{1}{\xi^2} d\sum_{x<d\le\xi} b(d) = \lim_{N\to\infty}[O(1/N) - 2\int_x^N \xi^{-3}\sum_{x<d\le\xi} b(d)d\xi] = 2\int_x^\infty \xi^{-3}O(\xi)d\xi = O(1/x)$。④ $\sum_{d>x} f(d)\log d = O(\log x/x)$。pf: LHS= $\sum_{d>x} b(d)\log d\, d^{-2} = \lim_{N\to\infty}\sum_{x<d\le N} b(d)\frac{\log d}{d^2} = \lim_{N\to\infty}\int_x^N \frac{\log\xi}{\xi^2} d\sum_{x<d\le\xi} b(d) = O(\log N/N) - \int_x^N \frac{1-\log\xi}{\xi^3}\sum_{x<d<\xi} b(d)d\xi = 2\int_x^\infty \frac{\log\xi - 1}{\xi^3}O(\xi)d\xi = O(\log x/x)$ 注意 $\log x + 1/x$ 导数是 $-\log x/x^2$，故得证

Exercise 3. 对 $x > 3$, $\sum_{1<n\le x} \frac{1}{n\log n} = \log\log x + C_1 + O(1/(x\log x))$. pf: Euler–Maclaurin, LHS= $\int_2^x \frac{d\xi}{\xi\log\xi} + \int_2^x(\{\xi\} - \frac12)(\frac{1}{\xi\log\xi})'d\xi + (\{x\} - \frac12)\frac{1}{x\log x} - (\{2\} - \frac12)\frac{1}{2\log 2} = \{\log\log x\} - \{\log\log 2 + \frac{1}{4\log 2} + \int_2^\infty[\{\xi\} - \frac12](\frac{1}{\xi\log\xi})'d\xi\} - \{\int_x^\infty[\{\xi\} - \frac12](\frac{1}{\xi\log\xi})'d\xi + (\{x\} - \frac12)\frac{1}{x\log x}\}$，即为所求。

Exercise 4. $r_2(n) := \#\{(a,b) \in \mathbb{Z}^2 \mid a^2 + b^2 = n\}$，求证 $\sum_{n\le x} r_2(n) = \pi x + O(\sqrt{x})$. pf: $\sum_{n\le x} r_2(n) = 4\sum_{0\le a\le\sqrt{x}}\sum_{0<b<\sqrt{x-a^2}} 1 + 1 = 4\sum_{0\le a\le\sqrt{x}}[\sqrt{x-a^2}] + 1 = 4\sum_{0\le a\le\sqrt{x}}\sqrt{x-a^2} + O(\sqrt{x})$。关于 a 单调，$\sum_{0\le a\le\sqrt{x}}\sqrt{x-a^2} = \int_0^{\sqrt{x}}\sqrt{x-a^2}\,da + O(\sqrt{x}) = -\frac{x\arccos(a/\sqrt{x})}{2}\Big|_0^{\sqrt{x}} + \frac{a\sqrt{x-a^2}}{2}\Big|_0^{\sqrt{x}} + O(\sqrt{x}) = \frac{\pi}{4}x + O(\sqrt{x})$，且 $a = 0$ 值也为 $O(\sqrt{x})$。

Exercise 5. $Q_2(x) = \#\{n \le x \mid n \text{ 无平方因子}\}$，则 $Q_2(x) = C_2 x + O(\sqrt{x})$ $(C_2 = 6/\pi^2)$. pf: $Q_2(x) = \sum_{n\le x}\mu^2(n) = \sum_{n\le x}\sum_{d^2|n}\mu(d) = \sum_{d\le\sqrt{x}}[x/d^2]\mu(d) = \sum_{d\le\sqrt{x}}\frac{x\mu(d)}{d^2} + O(\sqrt{x}) = x\sum_{d\ge1}\frac{\mu(d)}{d^2} + x\sum_{d>\sqrt{x}}\frac{\mu(d)}{d^2} + O(\sqrt{x}) = \frac{1}{\zeta(2)}x + O(x\int_{\sqrt{x}}^\infty \xi^{-2}d\xi) + O(\sqrt{x}) = \frac{1}{\zeta(2)}x + O(\sqrt{x})$。

Exercise 6. $\forall\varepsilon > 0, \exists C_\varepsilon > 0, \tau(n) < C_\varepsilon n^\varepsilon$. pf: 选 P s.t. $\forall p > P, \forall\alpha \in \mathbb{N}_{\ge1}, \alpha+1 < p^{\varepsilon\alpha}$，再取 $C_\varepsilon = \prod_{p<P}\sup_{\alpha\ge1}(\alpha+1)/p^{\alpha\varepsilon} < \infty$，则 $\tau(\prod p_i^{\alpha_i}) = \prod(\alpha_i+1) \le C_\varepsilon\prod p_i^{\varepsilon\alpha_i}$。

Exercise Sheet 3

Exercise 1. 若 $\sum_{p\le x} f(p)\log p = (ax+b)\log x + cx + O(1)$ $(x \ge 2)$，则 $\exists B = B(f)$, $\sum_{p\le x} f(p) = ax + (a+c)(\frac{x}{\log x} + \int_2^x \frac{dt}{\log^2 t}) + b\log\log x + B + O(\frac{1}{\log x})$. pf: $A(x) := \sum_{p\le x} f(p)\log p$, LHS= $\sum_{2\le n\le x} f(n)\log n\,\delta_p(n)\frac{1}{\log n} \overset{\text{Abel}}{=} \frac{A(x)}{\log x} + \int_{3/2}^x A(\xi)\frac{1}{\xi\log^2\xi}\mathrm{d}\xi =$ [条件] $ax + b + c\frac{x}{\log x} + O(\frac{1}{\log x}) + \int_{3/2}^x(\frac{a\xi+b}{\xi\log\xi} + \frac{c}{\log^2\xi} + \frac{E(\xi)}{\xi\log^2\xi})\mathrm{d}\xi$，其中 $E(\xi) \ll 1$，注意 $\frac{1}{x\log x} = (\log\log x)'$ 且 $a\int_{3/2}^x \frac{\mathrm{d}\xi}{\log\xi} = a\frac{\xi}{\log\xi}\Big|_{3/2}^x + a\int_{3/2}^x \frac{\mathrm{d}\xi}{\log^2\xi}$，积分下限可调整后放入 B 中。

Exercise 2. 证明对 $n \ge 16$, $\sum_{p|n}\frac1p \ll \log\log\log n$. pf: $\sum_{p|n}\frac1p = \sum_{p|n,\,p\le\log n}\frac1p + \sum_{p|n,\,p>\log n}\frac1p \ll$ (by Mertens) $\log\log\log n + \sum_{\substack{p>\log n\\ p|n}}\frac1p$. 由于 $(\log n)^{\log n/\log\log n} = n$ 从而 $> \log n$ 素数个数不超过 $\frac{\log n}{\log\log n}$，于是 $\sum_{p>\log n,\,p|n}\frac1p \le \frac{1}{\log n}\frac{\log n}{\log\log n} \ll \log\log\log n$。

Exercise 3. 证明 $\sum_p \frac{1}{p(\log\log p)^\alpha}$，$\alpha > 1$ 收敛，$\alpha \le 1$ 发散. pf: 忽略前面的项，$= \lim_{N\to\infty} S_N$，其中 $S_N = \sum_{5<n\le N}\delta_p(n)\frac1n\frac{1}{(\log\log n)^\alpha} = \sum_{5<p\le N}\frac1p\frac{1}{(\log\log N)^\alpha} + \int_5^N(\sum_{5<p\le\xi}\frac1p)\frac{\alpha}{(\log\log\xi)^{\alpha+1}}\frac{\mathrm{d}\xi}{\xi\log\xi}$，而 $\sum_{p\le x}\frac1p \sim \log\log x$ 且 $\int_5^\infty \frac{\mathrm{d}x}{x\log x(\log\log x)^\alpha}$ 在 $\alpha > 1$ 收敛，$\alpha \le 1$ 发散（用换元 $u = \log\log x$，$\frac{\mathrm{d}x}{x\log x} = \mathrm{d}u$）即证。

Exercise 4. 设 $S(x) = \#\{n \le x \mid p \mid n \Rightarrow p \le \sqrt{n}\}$，证明 $S(x) = (1 - \log 2)x + O(x/\log x)$. pf: $S(x) = [x] - \sum_{2\le p\le x}\sum_{m\le x/p,\,m<p} 1 = [x] - \sum_{2\le p\le\sqrt{x}}\sum_{m<p} 1 - \sum_{\sqrt{x}<p\le x}\sum_{m\le x/p} 1 = [x] - \sum_{2\le p\le\sqrt{x}}(p-1) - \sum_{\sqrt{x}<p\le x}[\frac{x}{p}]$，而 $[x] = x + O(1)$，$\sum_{2\le p\le\sqrt{x}}(p-1) \le \sqrt{x}\pi(\sqrt{x}) = O(x/\log x)$，$\sum_{\sqrt{x}<p\le x}[\frac{x}{p}] = x\sum_{\sqrt{x}<p\le x}\frac1p + O(\pi(x)) = x[(\log\log x + C + O(\frac{1}{\log x})) - (\log\log\sqrt{x} + C + O(\frac{1}{\log\sqrt{x}}))] + O(x/\log x) = x\log 2 + O(x/\log x)$，代回即得。

Exercise 5. 若 $\sum_{n\le x}\frac{\mu(n)}{n} = o(1)$，则 PNT. pf: $\sum_{1/2<n\le x}\mu(n) = \sum_{1/2<n\le x}\frac{\mu(n)}{n}n \overset{\text{Abel}}{=} x\sum_{1/2<n\le x}\frac{\mu(n)}{n} - \int_{1/2}^x\sum_{1/2<n\le\xi}\frac{\mu(n)}{n}\mathrm{d}\xi = o(x) + \int_{1/2}^{\sqrt{x}} M\,\mathrm{d}\xi + \int_{\sqrt{x}}^x o(1)\,\mathrm{d}\xi = o(x) \Rightarrow$ PNT。

Exercise 6. $L(N) = \operatorname{lcm}\{1, 2, \ldots, N\}$，求证 PNT$\Leftrightarrow \lim_{N\to\infty} L(N)^{1/N} = e$. pf: $\log L(N) = \sum_{p^\alpha\le N}\log p = \psi(N)$，故 $L(N)^{1/N} \to e$ iff $\frac{\psi(N)}{N} \to 1$ iff $\psi(x) \sim x$ iff PNT。

Exercise 7. $\forall n > 1$, $\exists p$ s.t. $p \| n!$. pf: Bertrand's 假设，取 $\frac{n}{2} < p < n$，则 $2p > n$，之后显然。

Exercise Sheet 4

Exercise 1. TFAE: (a) $f(n) \ll n^A$ $(A > 0)$; (b) $\sigma_a < +\infty$. pf: ① (a) 蕴含 $|f(n)| \le Cn^A$，故 $|L_f(s)| \le \sum_{n\ge1}\frac{C}{n^{\sigma-A}}$，即 $\operatorname{Re}(s) > A+1$ 时 $L_f(s)$ 绝对收敛，那 $\sigma_a < +\infty$。② (b) 蕴含 $\exists c > 0$，$\sum_{n\ge1}\frac{f(n)}{n^c}$ 绝对收敛，那 $\lim_{n\to\infty}\frac{|f(n)|}{n^c} = 0$，则 $f(n) \ll n^c$。

Exercise 2. (a) 找例子 s.t. $\sigma_a = \sigma_c + 1$; (b) 任给实数 σ_a, σ_c，$\sigma_a - 1 \le \sigma_c \le \sigma_a$，求 f 使 L_f 对应参数为 σ_a, σ_c. 对 (a) $f(n) = (-1)^n$ 有 $\sigma_c = 0, \sigma_a = 1$; (b) $f(n) = (-1)^n n^{\sigma_a - 1} + 1_{n=2^k}n^{\sigma_c}$，前一个对应 L 级数参数为 $\sigma_a - 1, \sigma_a$，后一个为 σ_c。

Exercise 3. 找 $F(s) = \sum_{m_1,m_2,m_3\ge1}(m_1+m_2+m_3)^{-s}$ 的绝对收敛横坐标，用 $\zeta(s)$ 表示. $F(s) = \sum_{n\ge1}\sum_{m_1+m_2+m_3=n} n^{-s} = \sum_{n\ge3}\binom{n-1}{2}n^{-s}$，$\binom{n-1}{2} \asymp n^2/2$，故 $\sigma_a = 3$。$F(s) = \sum_{n\ge3}\frac{n^2-3n+2}{2}n^{-s} = \frac12(\zeta(s-2) - 3\zeta(s-1) + 2\zeta(s))$ $(n = 0, 1, 2$ 项抵消)。

Exercise 4. $\chi(n)$ 定义为 $\chi(1) = 1$，$\chi(p_1^{\alpha_1}\cdots p_k^{\alpha_k}) = \alpha_1\alpha_2\cdots\alpha_k$，计算 (a) $\sum_{n\ge1}\frac{\tau(n^2)}{n^s}$; (b) $\sum_{n\ge1}\frac{2^{\omega(n)}}{n^s}$; (c) $\sum_{n\ge1}\frac{(-1)^{\Omega(n)}}{n^s}$; (d) $\sum_{n\ge1}\frac{3^{\omega(n)}\chi(n)}{n^s}$. pf: (a) $\tau(n^2)$ 可乘，用 Euler 乘积，$\sum_{n\ge1}\frac{\tau(n^2)}{n^s} = \prod_p(1 + \sum_{\nu\ge1}\frac{\tau(p^{2\nu})}{p^{\nu s}}) = \prod_p(1 + \sum_{\nu\ge1}\frac{2\nu+1}{p^{\nu s}}) = \prod_p((1-p^{-s})^{-1}(1-p^{-s})^{-1}(1+p^{-s})) = \prod_p(1-p^{-s})^{-3}(1-p^{-2s}) = \frac{\zeta(s)^3}{\zeta(2s)}$。(b) $2^{\omega(n)}$ 可乘，原式 $= \prod_p(1 + \sum_{\nu\ge1}\frac{2}{p^{\nu s}}) = \prod_p(1-p^{-2s})\prod_p(1-p^{-s})^{-2} = \frac{\zeta(s)^2}{\zeta(2s)}$。(c) 显然 $= \frac{\zeta(2s)}{\zeta(s)}$（是 Liouville 函数）。(d) 可乘，$= \prod_p(1 + \sum_{\nu\ge1}\frac{3\nu}{p^{\nu s}}) =$ (错位相减) $= \prod_p(1 + \frac{3p^s}{(p^s-1)^2}) = \prod_p(1-p^{-3s})(1-p^{-s})^{-3} = \frac{\zeta(s)^3}{\zeta(3s)}$。

Exercise 5. 对 $c > 0$, $\int_{(c)} = \int_{c-i\infty}^{c+i\infty}$. 证明 $(2\pi i)^{-1}\int_{(2)} y^s s^{-2}\,\mathrm{d}s = 0$ if $0 < y \le 1$, $= \log y$ if $y \ge 1$; 用包含 Λ 的和式表示 $(2\pi i)^{-1}\int_{(2)} y^s s^{-2}(-\zeta'/\zeta)(s)\,\mathrm{d}s$ *Solution.* 模仿讲义围道积分，模仿讲义 perron 公式可得 $(2\pi i)^{-1}\int_{(2)} y^s s^{-2}(-\zeta'/\zeta)(s)\,\mathrm{d}s = \sum_{n\le y}\Lambda(n)\log(y/n)$

Exercise 6. $\operatorname{Re}(s) > 1$, $\frac{1}{\zeta(s)} = s\int_1^\infty \frac{M(x)}{x^{s+1}}\,\mathrm{d}x$, $-\frac{\zeta'(s)}{\zeta(s)} = s\int_1^\infty\frac{\psi(x)}{x^{s+1}}\,\mathrm{d}x$，其中 $M(x) = \sum_{n\le x}\mu(n)$, $\psi(x) = \sum_{n\le x}\Lambda(n)$. pf: (a) $s\int_1^N\frac{M(x)}{x^{s+1}}\,\mathrm{d}x = s\sum_{n=1}^N\mu(n)\int_n^N\frac{1}{x^{s+1}}\,\mathrm{d}x = -[\sum_{n=1}^N\frac{\mu(n)}{N^s} - \sum_{n=1}^N\frac{\mu(n)}{n^s}]$ (取 $N \to \infty$) $= \sum_{n=1}^\infty\frac{\mu(n)}{n^s} = \frac{1}{\zeta(s)}$。(b) $s\int_1^N\frac{\psi(x)}{x^{s+1}}\,\mathrm{d}x = s\sum_{n=1}^N\Lambda(n)\int_n^N\frac{1}{x^{s+1}}\,\mathrm{d}x = -[\sum_{n=1}^N\frac{\Lambda(n)}{N^s} - \sum_{n=1}^N\frac{\Lambda(n)}{n^s}] \to \sum_{n=1}^\infty\frac{\Lambda(n)}{n^s} = -\frac{\zeta'(s)}{\zeta(s)}$。

Exercise Sheet 5

Exercise 1. $k = 2^\alpha$，$\forall a$ 奇，$a \equiv (-1)^{r_{-1}(a)}5^{r_0(a)} \pmod{2^\alpha}$，对 $\alpha \ge 2$，(a) $r_{-1}(a) \equiv \frac{a-1}{2} \pmod 2$; (b) $a \equiv a' \pmod{2^\alpha}$ iff $r_{-1}(a') \equiv r_{-1}(a) \pmod 2$，$r_0(a') \equiv r_0(a) \pmod{2^{\alpha-2}}$. pf: 显，见讲义。

Exercise 2. $\varphi^*(k)$ 是模 k 本原特征个数。求证 $\varphi^*(k) = \sum_{d|k}\mu(d)\varphi(k/d) = k\prod_{p\|k}(1 - \frac2p)\prod_{p^2|k}(1 - \frac1p)^2$. pf: 每个模 k 特征唯一由一个模 $d \mid k$ 本原特征诱导，故 $\varphi(k) = \sum_{d|k}\varphi^*(d) \Rightarrow \varphi^*(k) = \sum_{d|k}\mu(d)\varphi(k/d)$，第二个等式由可乘性，只需化 $k = p^\alpha$，$\sum_{d|p^\alpha}\mu(d)\varphi(p^\alpha/d) = \varphi(p^\alpha) - \varphi(p^{\alpha-1})$

Exercise 3. χ 模 k 的 D 特征，$k' \mid k$，TFAE: (a) χ 由 k' 本原特征诱导; (b) 若 $(n_i, k) = 1$ 且 $n_1 \equiv n_2 \pmod{k'}$，则 $\chi(n_1) = \chi(n_2)$; (c) $\chi(n_1) = 1$，$\forall n_1 \equiv 1 \pmod{k'}$，$(n_1, k) = 1$. pf: 讲义 6.3.2。

Exercise 4. $k = 2^\alpha$，$\alpha \ge 3$，$\chi(n, k, m_{-1}, m_0) = (-1)^{m_{-1}r_{-1}(n)}e(\frac{m_0 r_0(n)}{2^{\alpha-2}})$ $((n, 2) = 1)$，本原 iff $(m_0, 2) = 1$. pf: 见 6.3.6。

Exercise 5. 模 4 非主特征本原: $\chi_4(n) = (-1)^{\frac{n-1}{2}}$ $((n,2) = 1)$。模 8 本原实特征只有，$\chi_8(n) = (-1)^{\frac{n^2-1}{8}}$ $((n,2) = 1$，$\chi_4(n)\chi_8(n) = (-1)^{\frac{(n-1)(n+5)}{8}}$ $((n, 2) = 1)$。pf: 直接计算。

Exercise 6. $f : \mathbb{Z}/q\mathbb{Z} \to \mathbb{C}$, $\widehat{f} : \mathbb{Z}/q\mathbb{Z} \to \mathbb{C}$, $\widehat{f}(\alpha) := \frac{1}{\sqrt{q}}\sum_{a \bmod q} f(a)e^{2\pi i\alpha a/q}$。(a) $f, g : \mathbb{Z}/q\mathbb{Z} \to \mathbb{C}$，证明 $\sum_{a \bmod q} f(a)\overline{g(a)} = \sum_{\alpha \bmod q}\widehat{f}(\alpha)\overline{\widehat{g}(\alpha)}$; (b) $\sum_{a \bmod q}|f(a)|^2 = \sum_{a \bmod q}|\widehat{f}(a)|^2$; (c) $I = [A, B] \subset [0, q]$，计算 $\widehat{1_I}(\alpha)$; (d) 求证 $\sum_{\alpha \bmod q}|\widehat{1_I}(\alpha)| \ll \sqrt{q}\log q$; (e) 证明对本原 D-特征 $\chi \bmod q$，$\forall N \ge 1$，$\sum_{n=1}^N\chi(n) \ll \sqrt{q}\log q$. (a) pf: 换序求和，显，(用加法特征正交性。) (b) pf: (a) 中取 $f = g$。(c) pf: $\widehat{1_I}(\alpha) = \frac{1}{\sqrt{q}}\sum_{A\le a\le B} e^{2\pi i\alpha a/q}$ (等比级数)。不妨设 $A, B \in \mathbb{Z}$，(否则考虑 $[A]+1$, $[B]$; $\widehat{1_I}(0) = \frac{1}{\sqrt{q}}(B - A + 1)$, $\widehat{1_I}(\alpha) = \frac{1}{\sqrt{q}}\frac{e^{2\pi i\alpha A/q}(1 - e^{2\pi i\alpha(B-A+1)/q})}{1 - e^{2\pi i\alpha/q}}$。(d) pf: 其中 $L = \widehat{B} - \widehat{A} + 1$, LHS= $\frac{1}{\sqrt{q}}\sum_{\alpha \bmod q}\frac{|1 - e^{2\pi i\alpha L/q}|}{|1 - e^{2\pi i\alpha/q}|} \le \sum_\alpha\frac{1}{\sqrt{q}}\frac{2}{|1 - e^{2\pi i\alpha/q}|} = \frac{1}{\sqrt{q}}\sum_{\alpha \bmod q}\frac{1}{|\sin(\pi\alpha/q)|} \ll \frac{2}{\sqrt{q}}\sum_{\alpha=1}^{(q+1)/2}\frac{\pi}{2\alpha\pi/q} \ll \sqrt{q}\log q$，(利用 $x \in [0, \pi/2]$，$\sin x > \frac{2}{\pi}x$ 和 $\sin x = \sin(\pi - x)$。) (e) pf: 见讲义 thm 6.4.9。

Exercise 7. p 素数，$(\frac{n}{p})$ 是 D 特征模 p，求证模 p 的最小非二次剩余大小为 $O(\sqrt{p}\log p)$. pf: $\chi(n) := (\frac{n}{p})$ 不是模 1 本原特征，从而是模 p 本原特征。若 N 是最小的模 p 非二次剩余，则 $N - 1 = \sum_{n=1}^{N-1}\chi(n) \ll \sqrt{p}\log p$，故 $N \ll \sqrt{p}\log p$。

Exercise Sheet 6

Exercise 1. 有限 Abel 群 G: $G \simeq \widehat{G}$. pf: 见讲义 6.1 节。

Exercise 2. (a) $\operatorname{Re}(s) > 1$, $\zeta(s) = \frac{1}{s-1} + 1 - s\int_1^\infty\frac{\{\xi\}}{\xi^{s+1}}\,\mathrm{d}\xi$ (给出 $\operatorname{Re}(s) > 0$ 的亚纯延拓); (b) 洛朗展开 $\zeta(s) = \frac{1}{s-1} + \gamma + O(|s-1|)$ $(s \to 1)$, $\gamma = \lim_{n\to\infty}(\sum_{k=1}^n\frac1k - \log n)$. pf: (a) Euler-Maclaurin (讲义 2.2.3), $\zeta(s) = \lim_{N\to\infty}(1 + \sum_{1<n\le N}\frac{1}{n^s}) = 1 + \int_1^\infty\frac{1}{x^s}\,\mathrm{d}x - \int_1^\infty(\{x\} - \frac12)\frac{s}{x^{s+1}}\,\mathrm{d}x + \lim_{N\to\infty}\frac{\{N\} - 1/2}{N^s} - \frac{\{1\} - 1/2}{1^s} = 1 + \frac{1}{s-1} - s\int_1^\infty\frac{\{\xi\}}{\xi^{s+1}}\,\mathrm{d}\xi$。(b) $\zeta(s) = \frac{1}{s-1} + (1 - \int_1^\infty\frac{\{\xi\}}{\xi^2}\,\mathrm{d}\xi) + (\int_1^\infty\frac{\{\xi\}}{\xi^2}\,\mathrm{d}\xi - s\int_1^\infty\frac{\{\xi\}}{\xi^{s+1}}\,\mathrm{d}\xi)$，由讲义 2.2.4, $\gamma = 1 - \int_1^\infty\frac{\{\xi\}}{\xi^2}\,\mathrm{d}\xi$，而且 $\int_1^\infty\frac{\{\xi\}}{\xi^2}\,\mathrm{d}\xi - s\int_1^\infty\frac{\{\xi\}}{\xi^{s+1}}\,\mathrm{d}\xi = \int_1^\infty(\frac{\{\xi\}}{\xi^2} - \frac{\{\xi\}}{\xi^{s+1}})\,\mathrm{d}\xi - (s-1)\int_1^\infty\frac{\{\xi\}}{\xi^{s+1}}\,\mathrm{d}\xi = \int_1^\infty\frac{\{\xi\}}{\xi^{s+1}}(\xi^{s-1} - 1)\,\mathrm{d}\xi + O(s-1) = \int_1^\infty O(\frac{1}{\xi^{1+\delta}})O((s-1)\log\xi)\,\mathrm{d}\xi = O(s-1)$ $(s \to 1)$。

Exercise 3. $\prod_{p\le x}(1 - \frac1p)^{-1} = C\log x + O(1)$，求证 $C = e^\gamma$. pf: $P(x) = \prod_{p\le x}(1 - \frac1p)^{-1}$，$L = \log x$。Ex 6-2 (b) $\Rightarrow \zeta(1 + \frac1L) = L + O(1)$，而且 $\log\zeta(1 + \frac1L) = \sum_p -\log(1 - p^{-1-1/L}) = \sum_p\sum_{k\ge1}\frac{p^{-k(1+1/L)}}{k} = \sum_p -\log(1 - \frac1p)p^{-1/L} + o(1)$ $(x \to \infty)$。由 Abel，$\sum_p -\log(1 - \frac1p)p^{-1/L} = \int_{1-0}^\infty y^{-1/L}\,\mathrm{d}\sum_{2\le p\le y} -\log(1 - \frac1p) = \frac1L\int_2^\infty y^{-1/L-1}\sum_{2\le p\le y} -\log(1 - \frac1p)\,\mathrm{d}y$，故 $\log L + o(1) = \log\zeta(1 + \frac1L) = \frac1L\int_2^\infty y^{-1/L-1}\log\prod_{p\le y}(1 - \frac1p)^{-1}\,\mathrm{d}y + o(1)$。而 $C\log y + O(1) = C\log y(1 + O(\frac{1}{\log y}))$，故上式 $= \frac1L\int_2^\infty y^{-1/L-1}[\log C + \log\log y + O(\frac{1}{\log y})]\,\mathrm{d}y + o(1) = \log L + \log C - \gamma + o(1)$ (换元 $\frac{\mathrm{d}y}{y} = \mathrm{d}(\log y)$)，所以 $\log C = \gamma$，$C = e^\gamma$。

Exercise 4. $\chi \bmod q$，$\chi \ne \chi_0$，$x \ge 2$，证明 (a) $\sum_{n\le x}\frac{\chi(n)\log n}{n} = -L'(1,\chi) + O_\chi(\frac{\log x}{x})$; (b) $\sum_{n\le x}\frac{\chi(n)\log n}{n} = L(1,\chi)\sum_{m\le x}\frac{\chi(m)\Lambda(m)}{m} + O_\chi(\frac1x\sum_{n\le x}\Lambda(n))$，从而 $\sum_{n\le x}\frac{\chi(n)\Lambda(n)}{n} = O_\chi(1)$; (c) $\sum_{p\le x}\frac{\chi(p)\log p}{p} = O_\chi(1)$; (d) $\sum_{p\le x}\frac{\chi(p)}{p} = b(\chi) + O_\chi(\frac{1}{\log x})$，其中 $b(\chi) = \log L(1,\chi) - \sum_p\sum_{k\ge2}\frac{\chi(p^k)}{kp^k}$. pf: (a) $\sum_{n=1}^N\frac{\chi(n)}{n^s} = \frac{S(N)}{N^s} + s\int_1^N\frac{S(y)}{y^{s+1}}\,\mathrm{d}y$, $S(y) = \sum_{n\le y}\chi(n)$ 有界，故 L_χ 的 $\sigma_c \le 0$。讲义 4.10 $\Rightarrow -L'(1,\chi) = \sum_{n\ge1}\frac{\chi(n)\log n}{n} = \sum_{n\le x}\frac{\chi(n)\log n}{n} + \sum_{n>x}\frac{\chi(n)\log n}{n}$，而 $\sum_{n>x}\frac{\log n}{n}\chi(n) \overset{\text{Abel}}{=} \lim_{N\to\infty}[\sum_{x<n\le N}\chi(n)\frac{\log N}{N} - \int_x^N\sum_{x<n\le\xi}\chi(n)(\frac{\log\xi}{\xi})'\,\mathrm{d}\xi] = O_\chi(\frac{\log x}{x})$。(b) $\log = \Lambda * 1$，$\sum_{n\le x}\frac{\chi(n)\log n}{n} = \sum_{n\le x}\sum_{d|n}\frac{\chi(d)\Lambda(d)}{d}\frac{\chi(n/d)}{n/d} = \sum_{d\le x}\frac{\chi(d)\Lambda(d)}{d}\sum_{n\le x/d}\frac{\chi(n)}{n} = L(1,\chi)\sum_{d\le x}\frac{\chi(d)\Lambda(d)}{d} - \sum_{d\le x}\frac{\chi(d)\Lambda(d)}{d}\sum_{m>x/d}\frac{\chi(m)}{m}$，且 $\sum_{d\le x}\frac{\chi(d)\Lambda(d)}{d}\sum_{m>x/d}\frac{\chi(m)}{m} = (dm = e)\sum_{d\le x}\Lambda(d)\sum_{\substack{e>x\\ d|e}}\frac{\chi(e)}{e}$，Abel 求和 $\Rightarrow O_\chi(\frac1x\sum_{n\le x}\Lambda(n))$。(c) $\sum_{p\le x}\frac{\chi(p)\log p}{p} = \sum_{n\le x}\frac{\chi(n)\Lambda(n)}{n} - \sum_{\substack{p^\ell\le x\\ \ell\ge2}}\frac{\chi(p)^\ell\log p}{p^\ell}$，由 (b) 第一项有界，第二项 $\le \sum_{p\le\sqrt{x}}\log p\sum_{\ell\ge2}\frac{1}{p^\ell} = \sum_{p\le\sqrt{x}}\frac{\log p}{p(p-1)} = O(\sum\frac{1}{n^{1.5}})$ 有界。(d) $\sum_p\sum_{k\ge1}\frac{\chi(p^k)}{kp^k} = -\sum_p\log(1 - \frac{\chi(p)}{p}) = \log L(1,\chi)$，故 $\sum_p\frac{\chi(p)}{p} = \log L(1,\chi) - \sum_p\sum_{k\ge2}\frac{\chi(p^k)}{kp^k}$。而 $\sum_{p>x}\frac{\chi(p)}{p} = \sum_{p>x}\frac{\chi(p)\log p}{p}\frac{1}{\log p} \overset{\text{Abel}}{=} \sum_{x<p\le N}\frac{\chi(p)\log p}{p}\frac{1}{\log N} - \int_x^N(\frac{1}{\log\xi})'\sum_{x<p\le\xi}\frac{\chi(p)\log p}{p}\,\mathrm{d}\xi = O_\chi(\frac{1}{\log x})$，(由 (c))

Exercise 5. $|\arg s| \le \pi - \delta$，求证 $\Gamma'(s)/\Gamma(s) = \log s - \frac{1}{2s} + O_\delta(|s|^{-2})$. pf: $F(s) \triangleq \log\Gamma(s) - (s - \frac12)\log s + s - \frac{\log(2\pi)}{2}$。由 Stirling，$F(s) = O_\delta(|s|^{-1})$ $(s \to \infty$, $|\arg s| \le \pi - \frac\delta2)$。$|s|$ 充分大时，$\exists C_\delta > 0$，$D(s, C_\delta|s|) \subset \{|\arg z| \le \pi - \frac\delta2\}$，故 Cauchy 积分公式 $\Rightarrow |F'(s)| \le \frac{1}{C_\delta|s|}\sup_{|z-s|=C_\delta|s|}|F(z)| = O_\delta(\frac{1}{|s|^2})$。故 $F'(s) = O_\delta(\frac{1}{|s|^2})$，即为所求。

Exercise Sheet 7

引理. f 整函数，非多项式，$f(z) = \sum_{n=0}^\infty a_n z^n$，则 f 阶为 $\rho(f) = \limsup_{n\to\infty}\frac{n\log n}{-\log|a_n|}$ ($a_n = 0$ 时对应项视为 0)。pf: Cauchy 积分公式 $\Rightarrow |a_n| \le \sup_{|z|=r}|f(z)|\frac{1}{r^n}$。若 $\sup_{|z|=r}|f(z)| \ll \exp(r^{\rho+\epsilon})$，取 $r \asymp n^{1/(\rho+\epsilon)}$，则 $-\log|a_n| \gg \frac{n}{\rho+\epsilon}\log n + O(n)$，故 $\limsup_{n\to\infty}\frac{n\log n}{-\log|a_n|} \le \rho + \epsilon$。另一方面，若 $-\log|a_n| \gg \frac{n\log n}{\mu}$，则 $|a_n|r^n \le \exp(n\log r - \frac{n\log n}{\mu})$，$n \sim r^\mu \Rightarrow \sup_{|z|=r} f(z) \le \exp(Cr^\mu)$ (r 充分大)，$\Rightarrow \rho(f) \le \mu$，$\mu \to \limsup_{n\to\infty}\frac{n\log n}{-\log|a_n|}$。

Exercise 1. $\forall\alpha > 0$, $f_\alpha(z) = \sum_{j=1}^\infty\frac{z^j}{\Gamma(1+j/\alpha)}$ 为 α 阶整函数. pf: $a_j \triangleq \Gamma(1+j/\alpha)^{-1}$, Stirling $\Rightarrow -\frac1j\log|a_j| = \frac1\alpha(\log j + O(1)) \to +\infty$，故收敛半径 $R = \frac{1}{\limsup_{j\to\infty}|a_j|^{1/j}} = +\infty$，故整。$\limsup_{n\to\infty}\frac{n\log n}{-\log|a_n|} = \alpha$，故阶为 α。

Exercise 2. $f(z) \triangleq \sum_{n=0}^\infty e^{-n^2}z^n$ 是非多项式 0 阶整函数. pf: 因为 $\limsup_{n\to\infty}\frac{n\log n}{-\log|e^{-n^2}|} = 0$。

Exercise 3. 整函数 f 阶为 α，$f(0) \ne 0$，$\forall\epsilon > 0$，$\sum_{\rho,\,f(\rho)=0}\frac{1}{|\rho|^{\alpha+\epsilon}} < \infty$ (计重数). pf: Thm 10.2.6 说 $\sum_{|\rho|<R} 1 \ll R^{\alpha+\delta}$。设零点最小模长为 r，则 $\sum_\rho\frac{1}{|\rho|^{\alpha+\epsilon}} = \sum_{n>r}\sum_{2^n<|\rho|\le2^{n+1}}\frac{1}{|\rho|^{\alpha+\epsilon}} \ll \sum_{n>r}\frac{\sum_{|\rho|\le2^{n+1}} 1}{2^{n(\alpha+\epsilon)}} \le \sum_{n>r}\frac{2^{(n+1)(\alpha+\delta)}}{2^{n(\alpha+\epsilon)}} = \sum_{n>r}2^{\alpha+\delta-n(\epsilon-\delta)} < \infty$ (等比级数，一开始取 $\delta < \epsilon$)。

Exercise 4. Hadamard $\zeta^*(s) = e^{Bs}\prod_\rho(1 - \frac{s}{\rho})e^{s/\rho}$ 中 $B = -\sum_\rho\frac1\rho = -2\sum_{\gamma>0}\frac{\beta}{\beta^2+\gamma^2}$，其中 $\rho = \beta + i\gamma$，$\sum_\rho\frac1\rho \triangleq \lim_{x\to\infty}\sum_{|\rho|<x}\frac1\rho$. *Solution.* pf: $\frac{\zeta^{*\prime}(s)}{\zeta^*(s)} = B + \sum_\rho(\frac{1}{s-\rho} + \frac1\rho)$。$s \to 0 \Rightarrow \frac{\zeta^{*\prime}}{\zeta^*}(0) = B$。由 $\zeta^*(s) = \zeta^*(1-s)$，$\frac{\zeta^{*\prime}}{\zeta^*}(0) = -\frac{\zeta^{*\prime}}{\zeta^*}(1) = -[B + \sum_\rho(\frac{1}{1-\rho} + \frac1\rho)]$。由 ρ 零点，$1 - \rho$ 也为 ζ^* 零点，故对称主值下 $\sum_\rho\frac{1}{1-\rho} = \sum_\rho\frac1\rho$，得 $B = -\sum_\rho\frac1\rho$。又 $\rho = \beta + i\gamma$ 是零点时 $\bar\rho = \beta - i\gamma$ 也是零点，故 $B = -\sum_{\gamma>0}(\frac{1}{\beta+i\gamma} + \frac{1}{\beta-i\gamma}) = -2\sum_{\gamma>0}\frac{\beta}{\beta^2+\gamma^2}$。

Exercise 5. 设 χ 是模 q 的本原 Dirichlet 特征，$\chi(-1) = 1$。证明 $L(s,\chi) = \sum_{n\ge1}\chi(n)n^{-s}$ 解析延拓到整个复平面，并满足 $\pi^{-(1-s)/2}q^{(1-s)/2}\Gamma(\frac{1-s}{2})L(1-s,\overline{\chi}) = \frac{\sqrt{q}}{\tau(\chi)}\pi^{-s/2}q^{s/2}\Gamma(\frac s2)L(s,\chi)$，其中 $\tau(\chi) := \sum_{a=1}^q\chi(a)e^{2\pi ia/q}$; 并说明 $\chi(-1) = -1$ 时哪里出问题. pf: 除 $q = 1$ 的主特征外，本原特征非主，以下看非主情形。定义 $\theta(x,\chi) := \sum_{n\in\mathbb{Z}}\chi(n)e^{-\pi n^2x/q}$。由 $\chi(-1) = 1$，有 $\theta(x,\chi) = 2\sum_{n\ge1}\chi(n)e^{-\pi n^2x/q}$。当 $\operatorname{Re}s > 1$ 时逐项积分，且 $\int_0^\infty e^{-\pi n^2x/q}x^{s/2-1}\,\mathrm{d}x = (q/\pi)^{s/2}\Gamma(\frac s2)n^{-s}$，故记 $I(s,\chi) := \int_0^\infty\theta(x,\chi)x^{s/2}\frac{\mathrm{d}x}{x}$，则 $I(s,\chi) = 2\pi^{-s/2}q^{s/2}\Gamma(\frac s2)L(s,\chi)$ (∗)。接着按 $n = a + mq$ 分解并对内层用 Poisson 求和: $\theta(x,\chi) = \sum_{a \bmod q}\chi(a)\sum_{m\in\mathbb{Z}}e^{-\pi(a+mq)^2x/q}$，而 $\sum_{m\in\mathbb{Z}}e^{-\pi(a+mq)^2x/q} = \frac1q\sqrt{\frac qx}\sum_{k\in\mathbb{Z}}e^{2\pi ika/q}e^{-\pi k^2/(qx)}$; 又 χ 本原给 $\sum_{a \bmod q}\chi(a)e^{2\pi ika/q} = \overline{\chi}(k)\tau(\chi)$，于是 $\theta(x,\chi) = \frac{\tau(\chi)}{\sqrt{qx}}\theta(\frac1x,\overline{\chi})$。由定义知 $x \to \infty$ 时 $\theta(x,\chi)$ 指数衰减，由变换式知 $x \to 0^+$ 时也足够衰减，故 $I(s,\chi)$ 全纯; 再由 (∗) 定义 $L(s,\chi) := \frac12\pi^{s/2}q^{-s/2}I(s,\chi)/\Gamma(\frac s2)$，得到解析延拓。对 $I(1-s,\overline{\chi})$ 作 $x = 1/y$，并用上面对 $\overline{\chi}$ 的变换式，得 $I(1-s,\overline{\chi}) = \frac{\sqrt{q}}{\tau(\chi)}I(s,\chi)$; 把 (∗) 分别用于 (s,χ) 和 $(1-s,\overline{\chi})$ 即得函数方程。若 $\chi(-1) = -1$，则 $\theta(x,\chi)$ 中 n 与 $-n$ 两项抵消，所以 $\theta(x,\chi) \equiv 0$，上面的积分公式不能恢复 $L(s,\chi)$; 应改用 $\theta_1(x,\chi) = \sum_{n\in\mathbb{Z}}n\chi(n)e^{-\pi n^2x/q}$，Gamma 因子改为 $\Gamma(\frac{s+1}{2})$。

Exercise Sheet 8

Exercise 1. 若存在 $C > 0$ 使 $\zeta(s)$ 任一零点 ρ 满足 $\operatorname{Re}\rho < 1 - C/\log(2 + |\operatorname{Im}\rho|)$，不用显式公式证明存在 $c > 0$ 使 $\psi(x) = \sum_{n\le x}\Lambda(n) = x + O(xe^{-c\sqrt{\log x}})$. pf: 只需看非整数 x，整数点由 Perron 半权公式或 $x \pm 0$ 逼近。记 $L = \log x$，$L_T = \log(T+2)$，取 $a = 1 + 1/L$，$b = 1 - C/(2\log T)$，$3 \le T \le x$。若 $|\gamma| \le T$，则零点 $\rho = \beta + i\gamma$ 满足 $\beta < 1 - C/\log(2 + |\gamma|) \le 1 - C/\log(2 + T) < b$ (小 T 吸入常数)，故矩形 $b \le \operatorname{Re}s \le a$，$|\operatorname{Im}s| \le T$ 中 $-\zeta'/\zeta$ 只有 $s = 1$ 的简单极点，且 $-\zeta'/\zeta(s) = 1/(s-1) + O(1)$。Perron 公式给 $\psi(x) = \frac{1}{2\pi i}\int_{a-iT}^{a+iT}(-\zeta'/\zeta(s))x^s s^{-1}\,\mathrm{d}s + O(xL^2/T)$。由柯西留数定理移线到 $\operatorname{Re}s = b$，穿过 $s = 1$ 的留数为 x。无零矩形中标准 Jensen–Cauchy 估计给 $-\zeta'/\zeta(s) \ll_C L_T^2$，故新竖线为 $O_C(x^bL_T^3)$，横边与截断误差为 $O_C(xL^2/T)$。于是 $\psi(x) = x + O_C(x^bL_T^3 + xL^2/T)$。取 $T = \exp(A\sqrt{L})$，$A = \sqrt{C/2}$，则 $x^b = x\exp(-CL/(2\log T)) = xe^{-A\sqrt{L}}$ 且 $x/T = xe^{-A\sqrt{L}}$; 吸收对数幂得 $\psi(x) = x + O(xe^{-c_1\sqrt{\log x}})$.

Exercise 2. 设 μ 为 Mobius 函数。用 Perron 公式证明存在 $c > 0$ 使 $\sum_{n\le x}\mu(n) = O(xe^{-c\sqrt{\log x}})$。可用事实: 若 $\zeta(\sigma + i\tau) \ne 0$ 对 $\sigma \ge 1 - c_0/\log(|\tau| + 2)$ 成立，则当 $\sigma \ge 1 - \frac{c_0}{2\log(|\tau|+2)}$ 时 $1/|\zeta(\sigma + i\tau)| \ll_{c_0} \log(|\tau| + 2)$. pf: 设无零区域为 $\zeta(\sigma + i\tau) \ne 0$，$\sigma \ge 1 - \kappa/\log(|\tau| + 2)$。由题中 fact，若 $\sigma \ge 1 - \kappa/(2\log(|\tau| + 2))$，则 $1/|\zeta(\sigma + i\tau)| \ll_\kappa \log(|\tau| + 2)$。记 $M(x) = \sum_{n\le x}\mu(n)$，$L = \log x$，$L_T = \log(T+2)$，取 $a = 1 + 1/L$，$b = 1 - \kappa/(2\log T)$。Perron 公式给 $M(x) = \frac{1}{2\pi i}\int_{a-iT}^{a+iT}x^s(s\zeta(s))^{-1}\,\mathrm{d}s + O(xL/T)$。移线到 $\operatorname{Re}s = b$; 因 $1/\zeta(s)$ 在矩形内全纯且 $s = 1$ 是 $1/\zeta$ 的零点而非极点，所以无留数。由 fact，竖线为 $O_\kappa(x^bL_T^2)$，横边与截断误差为 $O_\kappa(xL/T)$，故 $M(x) = O_\kappa(x^bL_T^2 + xL/T)$。取 $T = \exp(B\sqrt{L})$，$B = \sqrt{\kappa/2}$，则 $x^b = xe^{-B\sqrt{L}}$ 且 $x/T = xe^{-B\sqrt{L}}$，吸收对数幂得 $\sum_{n\le x}\mu(n) = O(xe^{-c_2\sqrt{\log x}})$。

Exercise 3. $N \in \mathbb{N}$, $\operatorname{Li}(x) = \frac{x}{\log x}\sum_{n=0}^{N-1}\frac{n!}{(\log x)^n} + O_N(\frac{x}{(\log x)^{N+1}})$. pf: $\operatorname{Li}(x) = \int_2^x\frac{1}{\log u}\,\mathrm{d}u$，反复分部积分得 $\frac{x}{\log x}\sum_{n=0}^{N-1}\frac{n!}{(\log x)^n} + \int_2^x\frac{\mathrm{d}u}{(\log u)^{N+1}}$。余项 $= \int_2^{\sqrt{x}}\frac{\mathrm{d}u}{(\log u)^{N+1}} + \int_{\sqrt{x}}^x\frac{\mathrm{d}u}{(\log u)^{N+1}} \le \frac{\sqrt{x}}{(\log 2)^{N+1}} + \frac{x - \sqrt{x}}{(\frac12\log x)^{N+1}} \ll_N \frac{x}{(\log x)^{N+1}}$.

Exercise 4. 记 $M(x) := \sum_{n\le x}\mu(n)$ 与 $\psi(x) := \sum_{n\le x}\Lambda(n)$。则在 $0 \le \theta \le 1$ 下，$M(x) = O_\varepsilon(x^{1-\theta+\varepsilon})$ 当且仅当 $\psi(x) = x + O_\varepsilon(x^{1-\theta+\varepsilon})$. pf: $\theta = 0$ 平凡，设 $0 < \theta \le 1$，置 $\alpha = 1 - \theta < 1$。先证两边都等价于 $\zeta(s) \ne 0$ 对 $\operatorname{Re}s > \alpha$。若 $M(x) = O_\varepsilon(x^{\alpha+\varepsilon})$，则 $\operatorname{Re}s > 1$ 时 $1/\zeta(s) = \sum\mu(n)n^{-s} = s\int_1^\infty M(t)t^{-s-1}\,\mathrm{d}t$; 对 $\sigma = \operatorname{Re}s > \alpha$ 取 $\eta > 0$ 使 $\alpha + \eta < \sigma$，被积函数为 $O(t^{\alpha+\eta-\sigma-1})$，故积分在 $\operatorname{Re}s > \alpha$ 局部一致收敛，给出 $1/\zeta$ 全纯延拓，于是 ζ 无零。若 $\psi(x) = x + O_\varepsilon(x^{\alpha+\varepsilon})$，写 $\psi = x + E$，则 $\operatorname{Re}s > 1$ 时 $-\zeta'/\zeta(s) = s\int_1^\infty\psi(t)t^{-s-1}\,\mathrm{d}t = \frac{s}{s-1} + s\int_1^\infty E(t)t^{-s-1}\,\mathrm{d}t$，同理后一积分在 $\operatorname{Re}s > \alpha$ 全纯，故 $-\zeta'/\zeta(s) - 1/(s-1)$ 全纯; 若 ζ 在此半平面有零点，则 $-\zeta'/\zeta$ 有极点，矛盾。反过来设 $\zeta(s) \ne 0$ 对 $\operatorname{Re}s > \alpha$。任取 $\varepsilon > 0$，取 $b = \alpha + \varepsilon/3$，$c = 1 + 1/\log x$，$T = x$。在 $\operatorname{Re}s \ge b$ 中用标准零点避开估计 $1/\zeta(s) \ll_\varepsilon (|t| + 2)^{\varepsilon/3}$ 与 $-\zeta'/\zeta(s) \ll_\varepsilon \log^2(|t| + 3)$。Perron 公式给 $M(x) = \frac{1}{2\pi i}\int_{c-iT}^{c+iT}\frac{x^s}{s\zeta(s)}\,\mathrm{d}s + O(\log x)$，移线到 $\operatorname{Re}s = b$，不穿过留数，新竖线 $\ll_\varepsilon x^b\int_0^T(t+2)^{\varepsilon/3}(t+1)^{-1}\,\mathrm{d}t = O_\varepsilon(x^{\alpha+\varepsilon})$，横边与截断误差同阶，故 $M(x) = O_\varepsilon(x^{\alpha+\varepsilon})$。同样 $\psi(x) = \frac{1}{2\pi i}\int_{c-iT}^{c+iT}(-\zeta'/\zeta(s))x^s s^{-1}\,\mathrm{d}s + O(\log^2 x)$，移线只穿过 $s = 1$，因 $-\zeta'/\zeta(s) = 1/(s-1) + O(1)$，留数为 $\operatorname{Res}_{s=1}((-\zeta'/\zeta(s))x^s/s) = x$; 余下竖线 $\ll_\varepsilon x^b\int_0^T\log^2(t+3)(t+1)^{-1}\,\mathrm{d}t = O_\varepsilon(x^{\alpha+\varepsilon})$，故 $\psi(x) = x + O_\varepsilon(x^{\alpha+\varepsilon})$。整数 x 处用 Perron 半权公式或 $x \pm 0$ 逼近即可，代回 $\alpha = 1 - \theta$ 得证。

命题 8.0.1 (柯西积分公式（Cauchy integral formula）) 设 $\Omega \subset \mathbb{C}$ 为区域，f 在 Ω 上全纯。若 $\overline{D(z_0, r)} \subset \Omega$，则对任意 $n \ge 0$，$f^{(n)}(z_0) = \frac{n!}{2\pi i}\int_{|\zeta-z_0|=r}\frac{f(\zeta)}{(\zeta-z_0)^{n+1}}\,\mathrm{d}\zeta$。

留数定理（residue theorem）. 设 γ 为正向简单闭曲线，f 在 γ 及其内部除有限个孤立奇点 $a_1, \ldots, a_m$ 外全纯，则 $\int_\gamma f(z)\,\mathrm{d}z = 2\pi i\sum_{j=1}^m \operatorname{Res}(f, a_j)$。